Robert Jenkinson Alvarez

+34 722 559 713 | robert.j.jenkinson.alvarez@bath.edu | arXiv | GitHub | Personal Website

EDUCATION

London School of Economics and Political Science (LSE)

MSc Financial Mathematics (Distinction) — *Sep 2023 – Jul 2024*

- Black-Scholes; Interest Rate/Credit Risk-Hull-White/Vasicek/HJM; Stochastic Processes-Ito; Fixed Income; Portfolio Management.
- Quant Algathon (Top 2): Developed a 3-layer LSTM forecasting BTC/ETH returns using unconventional data (sentiment), with 1.2 out-of-sample Sharpe, 54% directional accuracy and MAE of 0.005.
- Analysed CRSP, TRACE/FISD and OptionMetrics data across equities, credit and derivatives, generated 24.7% excess returns over S&P 500 with 1.0 Sharpe, constructed 30-day yield curves via Newton optimisation and reduced options-hedging MSE by 80.3%.
- Implemented implicit, explicit, and Crank-Nicolson PDE solvers, priced options via Monte Carlo (control/antithetic variates) over 10,000 simulations with 97% confidence intervals and simulated SDEs via Euler, analysing sensitivity, stability and convergence.

University of Bath

BSc Mathematics (First-Class Honours) — *Sep 2020 – Jul 2023*

- Probability: Markov Processes; Martingales; Stochastic Systems; Discrete Prob.; Machine Learning; Measure Theory; Statistics.
- Pure/Applied: Advanced Real/ Complex Analysis; Algebra; ODEs/Control Theory; PDEs/Vector Calculus; Numerical Analysis (97%).

QUANTITATIVE RESEARCH & TRADING EXPERIENCE

Quantitative Researcher

Cross-Asset Manifold Alpha Engine — *Sep 2024 – Present*

- Discovered implied-volatility/Dupire disagreement forecasts 20-day excess variance with 0.857 validation and 0.811 untouched-test rank IC (NW t=25.1); independent variance-edge and projected-energy signals achieved test ICs of 0.553 and 0.342.
- Architected an alpha engine transforming 3.39 million quotes across 8 equities into arbitrage-aware geometric states, measuring volatility disagreement, projected energy and surface tension to reveal relative-value structure missed by conventional features.
- Pioneered a foliation-structured HamJEPA modelling market regimes as latent leaves, learning forecastable within-regime evolution through symplectic SSL and using transverse motion as a regime-break coordinate under non-stationarity.

Arbitrage-Free Option Surfaces (arXiv:2512.01967) — *Aug 2025 – Nov 2025*

- Designed an arbitrage-free equity option surface for liquid options via a Chebyshev tensor basis in moneyness and maturity space.
- Formulated calibration as a large-scale convex QP, encoding no-arbitrage constraints as sparse linear operators on a collocation grid, with orthogonal projections achieving 98-99% inside-spread and <1% static no-arb violations in calm regimes.
- Pioneered a fog layer modelling noise as a 3D density, minimising a convex energy with graph-Laplacian smoothness and band potentials, relaxing conflicting quotes while keeping the global surface arbitrage-free with <1% violations.

Optiver

Derivatives Quant Trading & Market Making Trainee — *Oct 2023 – Nov 2023*

- Reduced daily PnL variance by 30% and 95% VaR by 25% via delta-hedging and gamma rebalancing (IOC orders) over 4 weeks.
- Developed Black-Scholes-based options market-making algorithms in a 3-person team, optimising bid-ask spreads by dynamically weighting liquidity tiers and order book depth in Optibook's live trading simulated exchange.
- Improved order placement with a priority queue architecture, slashing latency by 40% through pruning low-liquidity strike/expiry pairs and caching Greeks calculations in real time for high-frequency updates.

MACHINE LEARNING & MATHEMATICAL RESEARCH

Beyond Isotropy in JEPAs: HamJEPA (arXiv:2605.20107); NeurIPS 2026 under review — *Oct 2025 – Present*

- Improved LeJEPA/SIGReg top-1 (linear-probe/kNN-20) on CIFAR-100 by +10.6/+6.5 (80 epochs) and +3.52/+4.89 (30 epochs) and on ImageNet-100 +7.52/+4.82 (45 epochs), raising effective-rank proxy from 109.7 to 125.6 and 51.2 to 96.5 respectively.
- Built phase-space JEPA with symplectic leapfrog dynamics, flow prediction, separable Hamiltonian and non-isotropic regularizers.
- Established theory for structured priors in JEPAs, proving that under an H-energy budget the optimal embedding prior is a canonical Hamiltonian-Gaussian via minimax/maximum-entropy and deriving a closed-form price of isotropy bound.

Variational Rigidity of Minimising Foliations — *Jan 2026 – Present*

- Proved the one-dimensional and arbitrary-dimensional cases of a stated open rigidity problem concerning minimising foliations of periodic semilinear elliptic equations.
- Combined global foliation coordinates, variational dynamics and Hopf-Riccati rigidity in one dimension and extended the argument through calibrated identities, Runge approximation and Bloch/Fermi spectral separation.

SKILLS & INTERESTS

- **Technical Skills:** Python (NumPy, Pandas, PyTorch, TensorFlow, SciPy, QuantLib, OSQP), MATLAB, R, SQL, LaTeX.
- **Communication:** Tutored metric spaces & stochastic processes to 5 university students, improving pass rates by 10%.
- **Languages / Interests:** Native Spanish & English (CPE C2). Piano (ABRSM Grade 8), Giveaway chess (2220 Elo, top 100 worldwide).